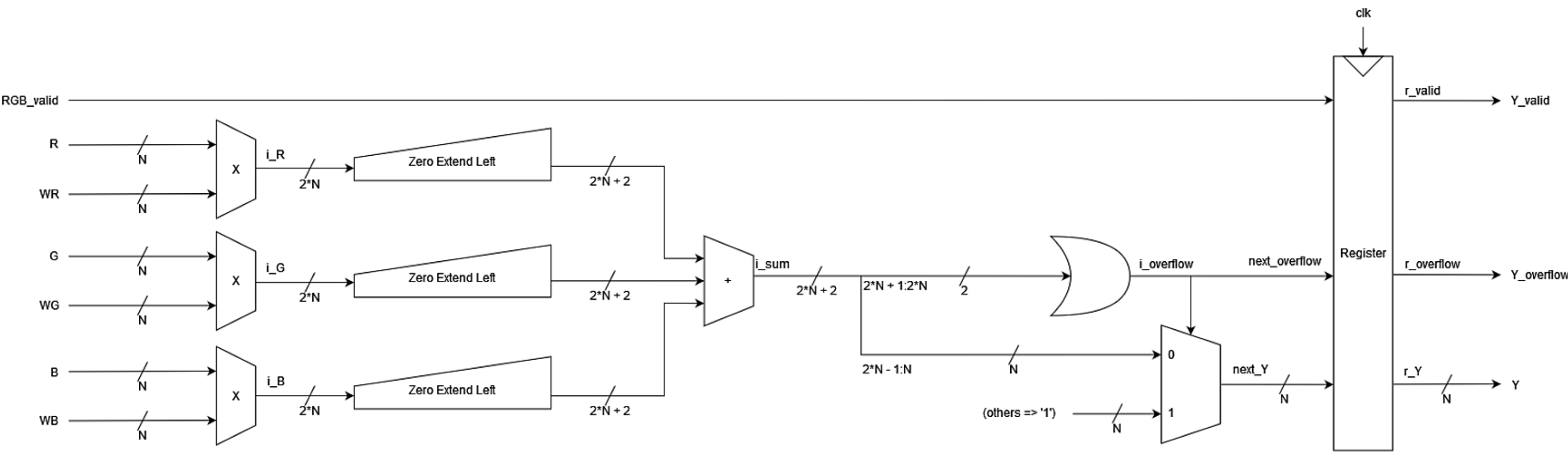


Task 1:

1. $a + b$ gives a maximum of 9 bits.
2. $a * b$ gives a maximum of 16 bits.
3. $a + b + c$ gives a maximum of 10 bits.
4. $(a + b + c + d) * e$ gives a maximum of 18 bits.

Non-pipelined:



Task 2:

c. $N + 2$ flip-flops are used.

e. $7*N + 3$ flip-flops are used.

Pipelined:

